



2 Real Analysis 2023

Tutorial 5

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The monotone convergence theorem

Problem 17 Prove that if a sequence is decreasing and bounded below, then it converges, and its limit is its infimum.

Solution.

Let $(a_n)_{n=1}^{\infty}$ be a decreasing bounded below sequence, and put $i = \inf_{n \in \mathbb{N}} a_n$. Let $\varepsilon > 0$. By Proposition 1.11, we can find $N \in \mathbb{N}$ such that $a_N < i + \varepsilon$. Since the sequence $(a_n)_{n=1}^{\infty}$ is monotone and i is its infimum, we have

$$i - \varepsilon < i \leq a_n \leq a_N < i + \varepsilon \quad \text{for all } n \geq N,$$

which implies

$$|i - a_n| < \varepsilon \quad \text{for all } n \geq N.$$

It follows that $\lim_{n \rightarrow \infty} a_n = i$.

An alternative solution: Consider the sequence $(-a_n)_{n=1}^{\infty}$. It is increasing and bounded above. Indeed, $a_n \geq a_{n+1}$ implies for the new sequence $-a_n \leq -a_{n+1}$, and if $a_n \geq m$ for all $n \in \mathbb{N}$, then $-a_n \leq -m$. By Theorem 2.17, $(-a_n)_{n=1}^{\infty}$ converges, and $\lim_{n \rightarrow \infty} (-a_n) = \sup_{n \in \mathbb{N}} (-a_n)$. It follows that $(a_n)_{n=1}^{\infty}$ converges, and $\lim_{n \rightarrow \infty} a_n = -\sup_{n \in \mathbb{N}} (-a_n) = \inf_{n \in \mathbb{N}} (a_n)$.

Problem 18 Let $(x_n)_{n=0}^{\infty}$ be a sequence defined recursively by

$$x_0 = 0, \quad x_n = \frac{1}{3}(2 + x_{n-1}^2), \quad n \in \mathbb{N}.$$

- (a) Show that $0 \leq x_n \leq 1$ for all $n = 0, 1, 2, \dots$
- (b) Show that the sequence $(x_n)_{n=0}^{\infty}$ is increasing.
- (c) Justify that the sequence $(x_n)_{n=0}^{\infty}$ converges.
- (d) Find $\lim_{n \rightarrow \infty} x_n$.

Solution.

- (a) We are going to use induction to show that $0 \leq x_n \leq 1$ for all $n = 0, 1, 2, \dots$

For the induction base, observe that $x_0 = 0$, so that $0 \leq x_0 \leq 1$.

For the induction step, assume that $0 \leq x_k \leq 1$ for some $k = 0, 1, 2, \dots$. We consider the following chain of equivalent inequalities:

$$\begin{aligned} 0 \leq x_k \leq 1 &\iff 0 \leq x_k^2 \leq 1 \\ &\iff 2 \leq 2 + x_k^2 \leq 3 \\ &\iff \frac{2}{3} \leq \underbrace{\frac{1}{3}(2 + x_k^2)}_{=x_{k+1}} \leq 1. \end{aligned}$$

Since $\frac{2}{3} > 0$, we see that $0 \leq x_{k+1} \leq 1$.

The statement follows by the principle of mathematical induction.

- (b) We will show by induction that $x_n \leq x_{n+1}$ for all $n = 0, 1, 2, \dots$.

For the induction base, observe that $x_0 = 0$ and $x_1 = \frac{2}{3}$, so that $x_0 \leq x_1$.

For the induction step, assume that $x_k \leq x_{k+1}$ for some $k = 0, 1, 2, \dots$. We have to show that $x_{k+1} \leq x_{k+2}$. Taking into account that all terms x_n are non-negative, we consider the following chain of equivalent inequalities:

$$\begin{aligned} x_k \leq x_{k+1} &\iff x_k^2 \leq x_{k+1}^2 \\ &\iff 2 + x_k^2 \leq 2 + x_{k+1}^2 \\ &\iff \underbrace{\frac{1}{3}(2 + x_k^2)}_{=x_{k+1}} \leq \underbrace{\frac{1}{3}(2 + x_{k+1}^2)}_{=x_{k+2}}. \end{aligned}$$

We see that $x_{k+1} \leq x_{k+2}$ as desired.

The statement follows by the principle of mathematical induction.

- (c) The sequence $(x_n)_{n=0}^\infty$ is increasing and bounded above. It converges by the Monotone Convergence Theorem.
- (d) Let $x = \lim_{n \rightarrow \infty} x_n$. Taking the limit of both sides of the formula $x_n = \frac{1}{3}(2 + x_{n-1}^2)$, we arrive at the following equation for x :

$$x = \frac{1}{3}(2 + x^2),$$

i.e.

$$x^2 - 3x + 2 = 0.$$

The solutions of this equation are $x = 1$ and $x = 2$. Since $x_n \leq 1$ for all $n = 0, 1, 2, \dots$, we must have $\lim_{n \rightarrow \infty} x_n \leq 1$. We conclude that $\lim_{n \rightarrow \infty} x_n = 1$.

Problem 19 We define the sequence $(u_n)_{n=1}^\infty$ recursively by

$$u_1 = \sqrt{2}, \quad u_{n+1} = \sqrt{2 + u_n}, \quad n \in \mathbb{N}.$$

- (a) Prove that the sequence $(u_n)_{n=1}^\infty$ converges.

- (b) Find $\lim_{n \rightarrow \infty} u_n$.

Solution.

- (a) We are going to prove by induction that the sequence $(u_n)_{n=1}^{\infty}$ is increasing and bounded above. We will show that

$$0 \leq u_n \leq u_{n+1} \leq 2 \quad \text{for all } n \in \mathbb{N}.$$

For the induction base, notice that $u_1 = \sqrt{2}$ and $u_2 = \sqrt{2 + \sqrt{2}}$. Using the fact that $0 < \sqrt{2} < 2$, we estimate

$$0 \leq \underbrace{\sqrt{2}}_{=u_1} \leq \underbrace{\sqrt{2 + \sqrt{2}}}_{=u_2} \leq \sqrt{2 + 2} = \sqrt{4} = 2.$$

For the induction step, assume that $0 \leq u_k \leq u_{k+1} \leq 2$ for some $k \in \mathbb{N}$. We have to prove that $0 \leq u_{k+1} \leq u_{k+2} \leq 2$. We consider the following chain of equivalent inequalities:

$$\begin{aligned} 0 \leq u_k \leq u_{k+1} \leq 2 &\iff 2 \leq 2 + u_k \leq 2 + u_{k+1} \leq 4 \\ &\iff \sqrt{2} \leq \underbrace{\sqrt{2 + u_k}}_{=u_{k+1}} \leq \underbrace{\sqrt{2 + u_{k+1}}}_{=u_{k+2}} \leq \underbrace{\sqrt{4}}_{=2}. \end{aligned}$$

Thus, $0 \leq u_{k+1} \leq u_{k+2} \leq 2$.

The statement follows by the principle of mathematical induction.

- (b) Let $u = \lim_{n \rightarrow \infty} u_n$. Taking the limit of both sides of the formula $u_{n+1} = \sqrt{2 + u_n}$, we arrive at the following equation for x :

$$u = \sqrt{2 + u}.$$

Taking the square of both sides, we arrive at

$$u^2 = 2 + u,$$

i.e.

$$u^2 - u - 2 = 0.$$

The solutions of this equation are $u = -1$ and $u = 2$. Since $u_n \geq 0$ for all $n \in \mathbb{N}$, we must have $\lim_{n \rightarrow \infty} u_n \geq 0$. We conclude that $\lim_{n \rightarrow \infty} u_n = 2$.

Series

Problem 20 For each of the series, find an expression for the partial sum. Determine whether the series converges or diverges. If the series converges, calculate its sum.

$$\begin{array}{lll}
\text{(a)} \sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n, & \text{(b)} \sum_{n=1}^{\infty} 2^{-n} 5^{n-1}, & \text{(c)} \sum_{n=0}^{\infty} \left(\frac{5}{n+2} - \frac{5}{n+3}\right), \\
\text{(d)} \sum_{n=1}^{\infty} \frac{1}{(n+3)(n+4)}, & \text{(e)} \sum_{n=0}^{\infty} \frac{2}{3^{2n+1}}, & \text{(f)} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + \sqrt{n+1}}.
\end{array}$$

Solution.

(a) This is a geometric series. We have

$$s_n = \sum_{k=0}^n \left(-\frac{2}{3}\right)^k = \frac{1 - \left(-\frac{2}{3}\right)^{n+1}}{1 - \left(-\frac{2}{3}\right)}.$$

Since $\left(-\frac{2}{3}\right)^{n+1} \rightarrow 0$ as $n \rightarrow \infty$, the series converges. Its sum is

$$\sum_{k=0}^{\infty} \left(-\frac{2}{3}\right)^k = \frac{1}{1 - \left(-\frac{2}{3}\right)} = \frac{3}{5}.$$

(b) This is again a geometric series. We have

$$s_n = \sum_{k=1}^n 2^{-k} 5^{k-1} = \frac{1}{5} \sum_{k=1}^n \left(\frac{5}{2}\right)^k = \frac{1}{5} \cdot \frac{5}{2} \sum_{k=0}^{n-1} \left(\frac{5}{2}\right)^k = \frac{1}{2} \frac{1 - \left(\frac{5}{2}\right)^n}{1 - \frac{5}{2}}.$$

Since this sequence is unbounded, the series diverges.

(c) This is a telescoping series. We have

$$\begin{aligned}
s_n &= \sum_{k=0}^n \left(\frac{5}{k+2} - \frac{5}{k+3}\right) \\
&= \left(\frac{5}{2} - \frac{5}{3}\right) + \left(\frac{5}{3} - \frac{5}{4}\right) + \left(\frac{5}{4} - \frac{5}{5}\right) + \cdots + \left(\frac{5}{n+2} - \frac{5}{n+3}\right) \\
&= \frac{5}{2} - \frac{5}{n+3}.
\end{aligned}$$

The series converges, and $\sum_{k=0}^{\infty} \left(\frac{5}{k+2} - \frac{5}{k+3}\right) = \frac{5}{2}$.

(d) We have

$$\frac{1}{(k+3)(k+4)} = \frac{1}{k+3} - \frac{1}{k+4}.$$

Thus, this is again a telescopic series. For its partial sums we have

$$\begin{aligned}
s_n &= \sum_{k=1}^n \frac{1}{(k+3)(k+4)} = \sum_{k=1}^n \left(\frac{1}{k+3} - \frac{1}{k+4}\right) \\
&= \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{n+3} - \frac{1}{n+4}\right) \\
&= \frac{1}{4} - \frac{1}{n+4}.
\end{aligned}$$

The series converges, and $\sum_{k=1}^{\infty} \frac{1}{(k+3)(k+4)} = \frac{1}{4}$.

(e) This is again a geometric series. We have

$$s_n = \sum_{k=0}^n \frac{2}{3^{2k+1}} = \frac{2}{3} \sum_{k=0}^n \left(\frac{1}{9}\right)^k = \frac{2}{3} \frac{1 - \left(\frac{1}{9}\right)^{n+1}}{1 - \frac{1}{9}}.$$

The series converges, and

$$\sum_{k=0}^{\infty} \frac{2}{3^{2k+1}} = \frac{2}{3} \frac{1}{1 - \frac{1}{9}} = \frac{3}{4}.$$

(f) We have

$$\frac{1}{\sqrt{k} + \sqrt{k+1}} = \frac{\sqrt{k} - \sqrt{k+1}}{(\sqrt{k} + \sqrt{k+1})(\sqrt{k} - \sqrt{k+1})} = \frac{\sqrt{k} - \sqrt{k+1}}{k - (k+1)} = \sqrt{k+1} - \sqrt{k}.$$

Thus, we again have a telescopic series. For its partial sums we have

$$\begin{aligned} s_n &= \sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k+1}} = \sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) \\ &= (\sqrt{2} - \sqrt{1}) + (\sqrt{3} - \sqrt{2}) + (\sqrt{4} - \sqrt{3}) + \cdots + (\sqrt{n+1} - \sqrt{n}) \\ &= \sqrt{n+1} - 1. \end{aligned}$$

Since this sequence is unbounded, the series diverges.